

LESSON GUIDE

Technology Plus

Unit 5: Software

3.3.1 – Productivity Software

Lesson Overview:

In this lesson, students will explore productivity software and identify the proper usage of specific software.

Students will:

- Explain the purpose of productivity software.
- Differentiate between the types of productivity software based on their use.
- Evaluate the function of productivity software and document its functionality.

Guiding Question: What is productivity software and how does it improve common tasks?

Suggested Grade Levels: 8-12

Technology Needed: Yes

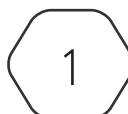
Approximate Time Required: 85-115 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 3.3 – Explain the purpose and proper use of software.
 - **Productivity software**
 - **Word processing**
 - **Spreadsheet**
 - **Presentation**
 - **Visual diagramming**

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Lesson Background

Background Information:

Productivity software is a category of programs that helps people create, manage, and share information efficiently. The four main types discussed in this lesson are word processing, spreadsheets, presentations, and visual diagramming. They are essential for personal, academic, and professional work. In this lesson, students will explore each software type, their purpose, and main uses. Students will also gain hands-on experience with free versions of this software and document the functionality of each type.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 3.3.1 – Productivity Software • Lab Slides 3.3.1 – Productivity Power Circuit
Materials per Student	• Student Notes 3.3.1 – Productivity Software • Student Handout 3.3.1 – Productivity Software • Assessment 3.3.1 – Productivity Software (optional)

Lesson Background

Cyber Terms:

Term	Definition
productivity software	software designed to help users complete tasks more efficiently by allowing for creating, organizing, and sharing work faster, easier, and more accurately
word processors	a program used to create, edit, and format text documents such as reports, letters, essays, and résumés
spreadsheet software	software that organizes, calculates, analyzes, and visualizes data using tables, charts, and formulas
presentation software	a tool for designing and delivering digital presentations or slideshows using visuals, text, and multimedia to make content more engaging and memorable
visual diagramming software	software used to create diagrams, flowcharts, and other visual representations to illustrate processes, hierarchies, and relationships

3.3.1 – Productivity Software

Guiding Question: What is productivity software and how does it improve common tasks?

Lesson Launch (5-10 minutes)

- **Ask:**

- What if you had to hand write every paper?
- What if you had to use graph paper and a calculator to create a budget?
- What if you had to create hand-drawn posters for each slide of a presentation?
- What if you had to use rulers, compasses, and stencils to create a diagram?

- Allow think time and let a few students share their answers.

Student answers should reflect that things would take a longer time, mistakes could happen, things would probably not be nice and neat, or easy to read or understand.

- **Say:** Without productivity software, mistakes mean we would have to start over, formatting would be manual and slow, collaboration would require mailing or physical meetings, and copies would have to be made on carbon paper or a copy machine. Productivity software doesn't just make these things easier, it completely changes the way we work, learn, and communicate. In this lesson you will explore productivity software and why it is so impactful in our daily lives.

Productivity Software: A Digital Toolbox (25-35 minutes)

- Distribute the Student Handout 3.3.1 – Productivity Software and have students fill in the answers during the lesson.
- Introduce the definition of **productivity software** and its benefits.
 - Before productivity software, creating projects was manual, slow, and prone to mistakes. Editing often meant restarting and collaboration meant passing the paper around instead of working together in real time.
 - Productivity software:
 - Saves time: fast editing, automation, templates
 - Improves accuracy and quality: spelling and grammar checks, data validation, and professional formatting
 - Boosts collaboration: multiple people can work on one doc, commenting and tracking changes
 - Increases flexibility: files can be updated and reused, multimedia support, work across devices
 - Enhances organization: files can be stored digitally, digital calendars and to-do lists
 - Tell students the productivity software you will discuss today is word processors, spreadsheets,

presentations, and visual diagramming.

- Talk about how people can use productivity software through paid versions, free versions, or mobile apps.
 - An example of commercial productivity software is Microsoft Office suite, it has applications for word processing, spreadsheets, presentations, and email.
 - An example of free productivity software is Google suite, it also has applications for word processing, spreadsheets, presentations, email, and more.
- Introduce **word processors**, its definition, uses, two examples, and a real-world scenario of the software in action and its features.
 - Word processors have existed on computers for a while but are becoming more advanced over time.
 - Word processors typically open with a blank page to type on but may also have options for templates.
 - Usage:
 - Creating Documents: Use word processors to create professional looking documents like reports, essays, and letters.
 - Formatting: Apply various text styles, such as bold or italics, adjust fonts and size, use headers and footers, and insert page numbers for a more polished appearance.
 - Editing: Utilize spell check, grammar check, and track changes features to ensure accuracy and clarity.
 - Collaboration: Use commenting and sharing features to collaborate with others on documents.
- Introduce **spreadsheet software**, its definition, uses, two examples, and a real-world scenario of the software in action and its features.
 - Spreadsheets allow users to search, sort, create formulas, and perform calculations on data that has been entered into its rows and columns.
 - Spreadsheets are often used for statistical analysis, financial calculations, and data management.
 - Usage:
 - Data Entry: Input and organize data in rows and columns.
 - Formulas: Apply mathematical and statistical formulas to perform calculations and analyze trends.
 - Charts and Graphs: Create visual representations of data to make trends and patterns more apparent.
 - Data Analysis: Use functions like pivot tables and data filters to summarize and analyze complex data sets.
- Introduce **presentation software**, its definition, uses, two examples, and a real-world scenario of the software in action and its features.
 - Presentations consist of individual pages shown on a screen for an audience while a presenter talks about a topic.
 - Presentations can include text, animations, pictures, charts, and data.

- Presentations are generally used for meetings, conferences, or educational purposes to help speakers demonstrate their ideas.
- Usage:
 - Slide Design: Develop visually engaging slides with text, images, and multimedia elements.
 - Presentation Flow: Arrange slides in a logical sequence to effectively communicate your message.
 - Visuals: Use diagrams, charts, and other graphics to illustrate points and enhance understanding.
 - Delivery: Practice and use presentation tools like speaker notes and transition effects to deliver a smooth and professional presentation.
- Introduce **visual diagramming software**, its definition, uses, two examples, and a real-world scenario of the software in action and its features.
 - Diagrams are useful for understanding the structure of a project or task and staying on the same page.
 - They are generally used for planning, brainstorming, and project management.
 - Usage:
 - Flowcharts: Design flowcharts to map out processes, workflows, and decision-making paths.
 - Organizational Charts: Illustrate hierarchies and relationships within an organization or project.
 - Mind Maps: Organize ideas and brainstorm by visually mapping out concepts and their connections.
 - Diagrams: Create various types of diagrams to visually represent information and facilitate understanding.

Lab: Productivity Power Circuit (45-60 minutes)

- This lab requires access to the CYBER.ORG Range, you will need to launch the Ubuntu Linux virtual machine, and Display Lab Slides 3.3.1 – Productivity Power Circuit and have students record their answers in the handout.
- Discuss the guiding questions and objectives for the lab.
- In this lab, students will access productivity software through the Ubuntu Linux virtual machine and complete simple tasks to test the functionality and benefits of the software.
- For each station, students will access different software and practice completing tasks.
- Each station will include the following slides:
 - An introduction to access the software
 - A scenario along with the task to be completed
 - Tips for using the software
 - A sample outline for what the final product could possibly look like
 - Documentation questions to be answered on the student handout

- Plan for each station to take about 15 minutes, this will allow time for students to complete the task and answer the questions on their handout.
- If you want students to submit their final products to you showing proof of participation, they could log into your LMS from the Range and upload files or log into a cloud storage application and upload their files through the Range.
- On the student handout, answers will vary based on student experience and utilization.

Putting It All Together

- Which software was the easiest to use? Why?

Students' answers will vary based on experience and opinion.

- Were there any tasks you completed that could be done in more than one program?

Accept all reasonable answers.

- How could you combine these tools for a bigger project?

Accept all reasonable answers.

Putting It All Together (10 minutes)

- Review what students learned today by discussing the questions below as a class.
- Think about a project you've done in school. Which type of productivity software would have helped you the most, and why?

Student answers will vary depending on the type of project and the productivity software they've used.

- How might your work or learning be different if you didn't have access to productivity software?

Student answers will vary, but they should connect that learning would be slower, more manual, and more time-consuming without productivity software.

- How do collaboration features in productivity software (like sharing, commenting, or real-time editing) change the way people work together compared to older methods?

Student answers will vary but should emphasize the ease, speed, and convenience of modern collaboration compared to older methods like passing paper copies or waiting on individual edits. Students may also recount experiences where sharing, commenting, or real-time editing could have been cumbersome depending on who or how many people were working on a document at once.

- If you had to choose just one type of productivity software to keep, which would you choose and what trade-offs would you face?

Student answers will vary based on which features they value most. They should also explain what challenges they would face without the other types of software.